

Problem Set 30

The Fundamental Theorem of Calculus, Part 2

1

(a) Find the following indefinite integrals, and check your answers by differentiating them.

i. $\int \cos x \, dx$

ii. $\int u^3 \, du$

iii. $\int \frac{x}{\sqrt[3]{x}} \, dx$

iv. $\int e^{-t} \, dt$

(b) Find the antiderivative $F(x)$ of $4 - \frac{3}{1+x^2}$ which satisfies $F(1) = 0$.

2

In February 2012, several people on campus got norovirus. Suppose $r(t) = 6 - 3\sqrt{t}$ gave the rate of change of the number of people on campus with norovirus, in people per day, where t is measured in days and $t = 0$ represents February 10. On February 14, 100 people on campus had norovirus.

(a) How many people on campus had norovirus on February 19?

(b) How many people on campus had norovirus on February 11?

(c) At what times between February 10 and February 20 was the number of people with norovirus increasing? At what times was it decreasing?

(d) Are your answers consistent with each other and with the fact that 100 people had norovirus on February 14?

3

(a) Find $\int_{-1}^1 |x| \, dx$ by interpreting it as a signed area.

(b) Show that $\frac{|x|^2}{2}$ is *not* an antiderivative of $|x|$ by showing that $\int_{-1}^1 |x| \, dx \neq \left. \frac{|x|^2}{2} \right|_{-1}^1$.

(c) To find an antiderivative of $|x|$, we use a strategy we've used before with the absolute value function:

we think about it piecewise. Find an antiderivative of $|x|$ on $(0, \infty)$. Then find an antiderivative of $|x|$ on $(-\infty, 0)$.

- (d) Fill in the blanks in the following statement (by putting together your answers from the previous part), and sketch a graph of the function $F(x)$:

One antiderivative of $|x|$ is

$$F(x) = \begin{cases} \underline{\hspace{2cm}} & \text{if } x > 0 \\ \underline{\hspace{2cm}} & \text{if } x < 0 \\ \underline{\hspace{2cm}} & \text{if } x = 0 \end{cases}$$